

Problem Set 4

Assignment Overview: This assignment is designed to help develop skills related to confidence intervals.

Instructions:

- This assignment is due Thursday, Nov. 6th at 11:59pm.
- Each individual must enter their MC answers in Canvas under Quizzes → PS4 MC. No hard copies will be accepted.
- You may re-submit your assignment as many times as you would like. Your last submission will be graded.
- Late assignments will receive at most half credit if they are turned in after the due date.
- You can work with anyone in this class on this assignment (but each individual must submit their own assignment).
- Read every possible answer. In some cases, an answer is a combination of other answers (for instance: “All of the above”).

Multiple Choice (1 point each)

Scenario 1: Suppose we take a random sample of 100 items and find a sample mean of 80. Further suppose that we know that the population variance is 225.

1. Refer to Scenario 1. Provide a 90% confidence interval for the population mean.
 - (A) 80 ± 2.4675
 - (B) 80 ± 2.94
 - (C) 80 ± 1.47
 - (D) 80 ± 1.1525
2. Refer to Scenario 1. Provide a 95% confidence interval for the population mean.
 - (A) 80 ± 2.4675
 - (B) 80 ± 2.94
 - (C) 80 ± 1.47
 - (D) 80 ± 1.1525
3. Refer to Scenario 1. Now assume that the sample mean was obtained from a sample of 400 items. Provide a 95% confidence interval for the population mean in this case.
 - (A) 80 ± 2.4675
 - (B) 80 ± 2.94
 - (C) 80 ± 1.47
 - (D) 80 ± 1.1525

For your consideration: What was the effect of the larger sample size on the interval estimate? What happened to the width of the confidence interval as the confidence level increased?

Scenario 2: Suppose a sample of 1000 is taken from a population with population proportion, $p = 0.40$.

4. Refer to Scenario 2. What is the expected value of $\hat{p}$?
 - (A) 0.40
 - (B) 0.60
 - (C) 400
 - (D) 1000
5. Refer to Scenario 2. What is the standard error of $\hat{p}$?
 - (A) 0.00249
 - (B) 0.00549
 - (C) 0.01549
 - (D) 0.03549
6. Refer to Scenario 2. What is the probability that the sample proportion will be within ± 0.05 of the population proportion?
 - (A) 0.9868
 - (B) 0.9874
 - (C) 0.9988
 - (D) 0.9998

For your consideration: Sketch the population distribution. Now sketch the sampling distribution of $\hat{p}$. Explain why we can use a normal distribution in Question 6, when the population distribution is clearly not normally distributed.

Scenario 3: Suppose you take 64 random draws from a distribution and find a sample mean of 35. Further suppose you know that the population standard deviation is 5.

7. Refer to Scenario 3. What is the standard error of the sample mean?
 - (A) 0.625
 - (B) 0.125
 - (C) 0.250
 - (D) 0.750
8. Refer to Scenario 3. At the 95% confidence level, what is the margin of error?
 - (A) 0.925
 - (B) 1.025
 - (C) 1.125
 - (D) 1.225
9. Refer to Scenario 3. Provide a 95% confidence interval for the population mean.
 - (A) [32.775, 37.225]
 - (B) [33.775, 36.225]
 - (C) [31.775, 38.225]
 - (D) [30.775, 39.225]
10. Refer to Scenario 3. Provide a 99% confidence interval for the population mean.
 - (A) 35 ± 1.61
 - (B) 35 ± 1.78
 - (C) 35 ± 1.96
 - (D) 35 ± 2.135

For your consideration: What would happen to the size of the confidence intervals if your sample size was larger?

11. Suppose you see that a 95% confidence interval (CI) is estimated to be 36 to 44. Further suppose that the population standard deviation is 15. What sample size was used in the estimation of the CI?
- (A) 54
 - (B) 64
 - (C) 128
 - (D) 256

Scenario 4: A study of attorney fees in Chapter 11 filings shows that the average attorney fee is \$3243.57. The sample size of the study is 1,001.

12. Refer to Scenario 4. Construct a 95% CI for the population mean. Assume that the standard deviation is known to be \$973.21.
- (A) 3243.57 ± 60.29
 - (B) 3243.57 ± 60.35
 - (C) 3243.57 ± 182.12
 - (D) 3243.57 ± 192.13
13. Refer to Scenario 4. Construct a 95% CI for the population mean. In this case, the standard deviation of the population is unknown. Assume that the standard deviation from the sample is \$973.21.
- (A) 3243.57 ± 60.29
 - (B) 3243.57 ± 60.35
 - (C) 3243.57 ± 182.12
 - (D) 3243.57 ± 192.13
14. Refer to Scenario 4. Suppose that the sample size is 101 instead of 1001. Construct a 95% CI using the information in Question 13.
- (A) 3243.57 ± 60.29
 - (B) 3243.57 ± 60.35
 - (C) 3243.57 ± 182.12
 - (D) 3243.57 ± 192.13

For your consideration: Why is your CI increasing in size in each part?

Scenario 5: For a t-distribution with 16 degrees of freedom, find the area (or probability), in each region:

15. To the right of 2.120

(A) 0.025

(B) 0.05

(C) 0.90

(D) 0.95

16. To the left of 1.337

(A) 0.025

(B) 0.05

(C) 0.90

(D) 0.95

17. To the left of -1.746

(A) 0.025

(B) 0.05

(C) 0.90

(D) 0.95

18. To the right of 2.583

(A) 0.01

(B) 0.02

(C) 0.95

(D) 0.97

19. Between -2.120 and 2.120

(A) 0.01

(B) 0.02

(C) 0.95

(D) 0.97

20. How large a sample should be drawn to provide a 90% confidence interval with a margin of error of 8? Assume $\sigma = 20$. Use the standard normal distribution (even though the sample size will be less than 30).

- (A) 17
- (B) 21
- (C) 25
- (D) 34

Scenario 6: These sampled data are from a population: 10, 8, 12, 15, 13, 11, 6, 15.

21. Refer to Scenario 6. What is the point estimate of the population mean?

- (A) 10.25
- (B) 11.25
- (C) 12.25
- (D) 13.25

22. Refer to Scenario 6. What is the point estimate of the population standard deviation?

- (A) 3.196
- (B) 3.296
- (C) 3.396
- (D) 3.496

23. Refer to Scenario 6. Using a 95% confidence level, what is the margin of error for the estimation of the population mean?

- (A) 2.672
- (B) 2.472
- (C) 2.272
- (D) 2.072

24. Refer to Scenario 6. What is the 95% CI for the population mean?

- (A) 11.25 ± 2.672
- (B) 11.25 ± 2.572
- (C) 11.25 ± 2.472
- (D) 11.25 ± 2.372